Wajih Habrah

Biomedical Engineer

Clinical engineering | Robotics | Digital twins

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Profile

Biomedical engineer with hospital experience at Södersjukhuset and an MSc in Biomedical Engineering from Chalmers. Combines medical-equipment maintenance, troubleshooting and technical documentation with hands-on development of robotic prototypes, embedded sensing and Unity-based digital twins. Practical project experience connecting hardware and software using ROS 2 and Python.

Professional experience

Biomedical Engineer Jan. 2018–Present

Södersjukhuset · Stockholm, Sweden

Full-time: Jan. 2018–Dec. 2021.
Part-time hourly/consulting assignments: Jan. 2022–Present.

- Performed preventive and corrective maintenance, calibration, functional checks and repairs on equipment used in intensive care, operating rooms and diagnostics, including patient monitoring, respiratory-care equipment and mobile C-arms.
- Improved existing preventive-maintenance documentation for Ziehm C-arms with illustrated, step-by-step instructions, making service procedures clearer, more repeatable and easier to trace.
- Documented service work, supported clinical staff with equipment issues, and contributed to technical investigations and follow-up of medical-equipment safety incidents.

Featured engineering project

Robotic Phantom Knee & Digital Twin 2025–2026

Master's thesis · Chalmers University of Technology

ROS 2 · Python · Unity / C# · Raspberry Pi · Blender

- Developed a one-degree-of-freedom robotic knee research prototype for early-stage exoskeleton testing, integrating antagonistic actuators, pneumatic soft-tissue simulation and embedded sensors.
- Implemented modular ROS 2 nodes in Python for motor and pneumatic control, sensor acquisition, parameter management, logging, emergency stop and controlled air release.
- Built a Unity digital twin with bidirectional ROS–TCP communication for command input, live state visualisation and monitoring of the physical prototype.
- Evaluated subsystems and partial-system integration, including data streaming, closed-loop pressure control and emergency-stop/air-release behaviour.

Project details and demonstration | Published thesis

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Education

MSc in Biomedical Engineering Sep. 2023–Jun. 2026

Chalmers University of Technology · Gothenburg, Sweden

Thesis: Robotic Phantom Knee and Digital Twin Platform for Early-Stage Exoskeleton Validation.

Selected study areas: Rehabilitation engineering; biomedical modelling and simulation; applied signal processing; deep machine learning; image analysis; MRI; digital health.

Bachelor's Degree in Biomedical Engineering 2007–2013

Damascus University · Damascus, Syria

Selected study areas: Biomechanics; control systems; medical electronics and biomedical measurements; medical devices; microprocessors; signal and image processing; prosthetics.

Technical skills

Clinical engineering Preventive and corrective maintenance, calibration, functional verification, fault investigation and service documentation.

Robotics and embedded systems ROS 2 (rclpy), closed-loop control, motor and pneumatic actuation, Raspberry Pi, microcontrollers, IMUs, pressure/temperature sensors and encoders.

Digital twins and 3D Unity, ROS-TCP-Connector, Blender, interactive visualisation, low-poly modelling, texture baking and preparation for 3D printing.

Programming Python, C#, C++, MATLAB.

Development tools Git, GitHub, GitLab, Unity Version Control, Microsoft Office.

Independent projects

Interactive 3D development 2019–Present

Personal projects · Unity, Blender and Firebase

- Develop interactive 3D prototypes and explore backend-supported dynamic configuration using Unity and Firebase.
- Create and optimise 3D assets in Blender for visualisation, texture baking, low-poly workflows and 3D printing.

Languages and additional information

Languages: Arabic (native), Swedish (fluent), English (fluent).

Citizenship: Swedish and Syrian. **Driving licence:** Swedish category B, since 2017.

Interests: Basketball, drawing, sculpture, vehicle troubleshooting and repair.